
AMR-X PROTOTYPE V1 · MECHANICAL ENGINEERING

Drive-Wheel Suspension

Options, Trade-offs, Recommendation and Procurement

2-drive / 4-caster differential AMR · 130 kg design load · 24 V · 0.8–1.5 m/s

List of Abbreviations

Abbreviation	Meaning
AMR	Autonomous Mobile Robot
AMR-X	Designation of this prototype platform (the robot studied in this report)
AGV	Automated Guided Vehicle
PU	Polyurethane (caster/wheel tyre material)
RFQ	Request for Quotation
SIL	Safety Integrity Level (functional-safety classification, IEC 61508)
PLd	Performance Level d (functional-safety rating, ISO 13849)
CoG	Centre of Gravity
EC	Electronically Commutated — brushless motor family (e.g. Maxon EC 45)
GP	Gearhead, Planetary — gearbox family (e.g. Maxon GP 42C)
USPTO	United States Patent and Trademark Office
IEEE	Institute of Electrical and Electronics Engineers
IROS	International Conference on Intelligent Robots and Systems
<i>Units</i>	
V	Volt
W	Watt
N	Newton
Nm / mNm	Newton-metre / milli-newton-metre
kg	Kilogram
mm	Millimetre
m/s	Metres per second

1. Why this question exists

The layout uses **six ground contacts**: two central drive wheels plus four corner casters. On a rigid frame that is one contact too many. Any three points define a plane, so a rigid six-contact base behaves like a four-legged table on an uneven floor: it rocks, and whichever contacts lift stop carrying load.

For a differential drive this is not cosmetic. Steering depends on *both* drive wheels being firmly pressed to the floor. If one unloads on an expansion-joint seam or a dock-plate lip, the wheel slips, the robot yaws unpredictably, odometry drifts, and — with our payload centre of gravity up at ≈ 398 mm — the rocking is amplified. The real question is therefore not “do we want a fancy suspension” but **how do we guarantee all six contacts stay loaded on a real warehouse floor**.

Design target. Production AMRs in this class (e.g. MiR) publish a traversable floor-gap tolerance of ≈ 30 mm at ≈ 27 mm ground clearance [7]. That confirms the target: small, repeated unevenness (joints, ramps, thresholds), *not* rough terrain.

2. Candidate mechanisms

Six approaches span the spectrum from doing nothing to full rover suspension.

Option	How it works	Traction	Complexity	Fit
A. Rigid	all wheels bolted hard to frame	Poor	Lowest	Flat floor only
B. Sprung casters	springs on the corner casters	Fair	Low	Light robots
C. Sprung drive units	each drive wheel on a pre-loaded spring/slide (~15–25 mm travel)	Good	Low–Med	Our case
D. Transverse rocker	both drive units on one central pivot beam (see-saw)	Good	Low	Our case
E. Independent arms	each drive wheel on its own sprung swing-arm	Very good	Med–High	Uneven floors
F. Rocker-bogie	linked rockers + body differential (Mars rover)	Excellent	High	Rough terrain (overkill)

The two that fit us are C and D. **C** (sprung drive units) directly guarantees a constant downward force on each drive wheel, so traction is maintained over seams and ramps and the payload is isolated from jolts. **D** (transverse rocker) is the simplest passive way to keep *both* drive wheels loaded: when one rises, the beam rocks and the other is pushed down by the same amount. In practice the two are often combined — a sprung drive pod mounted on a rocker.

3. Real-robot precedents

Option C — sprung drive wheels. The **iRobot Roomba** is the textbook case: each differential drive wheel is a self-contained spring-loaded module (motor + gearbox + spring) that compresses and extends independently to keep floor contact; the design rationale is exactly ours — losing drive-wheel contact loses steering accuracy [2, 3]. At our scale, AMR patent **US 12,128,554** places the drive units on coil springs and linear bearings with a ≈ 25 mm stroke and couples both wheels (“parallel” suspension) to keep the track width constant and stay narrow-aisle compact [4]. **MiR** AMRs use a central differential drive whose published gap/clearance figures only make sense with loaded/sprung drive wheels [7].

Option D — rocker / equalizer. The Mars-rover rocker-bogie is the most-illustrated example of the pivot principle: if one wheel goes up, the linked wheel is forced down by the same amount [6]. AMR driving-module patents use the same single-pivot “rocker” idea in a compact drive unit [5]. We need only the *single central rocker*, not the full bogie.

4. Recommendation: Solution C (sprung drive units)

We recommend Solution C — placing each drive wheel on its own pre-loaded spring suspension. It is the mechanism the field converges on for indoor differential AMRs because it attacks our exact failure mode (a drive wheel unloading) at the source, isolates the electronics and payload from floor shock, and — unlike a bare rocker — gives a known, repeatable wheel load that can be tuned to our 65 kg-per-wheel share. It is also the only one of the two available as a *finished, bolt-on product* (Section 6), which de-risks and accelerates a prototype.

4.1 The trade-off you must accept

Buying Solution C as a ready module means replacing the Maxon EC 45 flat + GP 42C. The off-the-shelf “AGV suspension drive wheel” is sold as a sealed assembly — wheel + gearbox + *motor* + encoder + spring (sometimes brake and controller) — all dimensioned around *its own* motor. There is no slot to insert the EC 45. So adopting a ready C-module retires our discrete motor/gearbox selection. The upside: these modules use 400 W-class motors, which *exceed* the 2×70 W Maxon set-up — so torque and power go *up*, not down (see Section 5–6).

If keeping the EC 45 + GP 42C is mandatory, Solution C must instead be *fabricated* (a sprung vertical linear slide per drive unit, built from catalogue linear guides + springs), or Solution D (a fabricated rocker) is used — both keep the Maxon and all dimensions, but neither is a single purchasable kit.

5. Specifications to preserve or improve

Any replacement module must **meet or beat** the current drivetrain. Targets derived from the sizing study:

Parameter	Current GP 42C)	(EC 45 + Requirement for a kit
Supply voltage	24 V	24 V (match battery)
Max speed	1.5 m/s (33:1 gives 2.85 top)	≥1.5 m/s
Continuous tractive force (pair)	≈69 N	≥69 N; ≥100 N desired for ramps
Continuous torque (per motor)	130 mNm [1]	meet/exceed at the wheel
Wheel diameter	Ø180 mm	Ø180 mm to preserve geometry
Per-wheel load (spring tuning)	≈65 kg	spring rated/tuned for ~65 kg/wheel
Encoder	yes (Hall / E6B2)	integrated encoder

A 400 W-class integrated drive wheel clears the speed, torque and power targets with margin (“better”); the two items that need active confirmation are **24 V** availability and a **spring rate tuned to our light 65 kg/wheel** (most modules are built for far heavier robots, so an untuned spring can be too stiff to deflect).

6. Ready-to-buy kits (Solution C) suited to our use case

The following provide drive + suspension in one unit and are appropriate (with the noted configuration) for a 130 kg indoor AMR. All are customisable with low minimum order quantity.

Product	Why it fits us / what to configure	Suspension
ZHLUN Robotic Drive Unit (electric differential wheel) [8]	Bundles motor, encoder, reducer, controller and PU wheel; two-stage damping (coil spring + PU block). Request 24 V, Ø180 wheel, ≥ 1.5 m/s, spring tuned to 65 kg/wheel.	Spring + PU block (built in)
TZBOT AGV Suspension Drive Wheel [9]	Dedicated suspension differential unit; select the <i>smallest</i> model (the TZCS heavy-load variants are 4-tonne class — avoid). Confirm 24 V and wheel Ø.	Spring suspension (built in)
BeUDMKE Shock-Absorbing Drive Wheel [10]	One-stop wheel + motor + driver + gearbox with shock absorption; customisable voltage, wheel Ø and ratio.	Shock-absorbing (built in)
ez-Wheel SWD 125 [11]	24 V integrated <i>safety</i> wheel drive purpose-built for AMRs (SIL2/PLd), encoder included — strong on functional safety. Note: Ø125 wheel (changes ride height) and confirm whether compliance is built in or added.	Integrated drive (verify)
Bosch Rexroth ROKIT Motor [12]	Premium plug-and-drive wheel (wheel, gearbox, brake, motor, encoder, controller). Pallet-scale — likely oversized; rigid, so pair with compliance. Reference-grade fallback.	Rigid (add compliance)

Primary picks for us: the **ZHLUN** and **BeUDMKE** suspension drive wheels, because they explicitly include the spring/PU compliance (true Solution C), are customisable to 24 V / Ø180 / ≥ 1.5 m/s, and carry far more than 65 kg so the motor specs land *above* ours. The single open point on both is spring tuning for our light per-wheel load — specify it in the RFQ.

7. Priority of this modification

Priority: MEDIUM now, rising to HIGH before any real-floor deployment.

- **Not a bring-up blocker.** On a flat lab/test floor a rigid mount runs; the robot can be powered up, driven and have its electronics, navigation and safety chain validated *before* the suspension is fitted. So it does not gate first motion.
- **Becomes critical for deployment.** On any production floor with expansion joints, dock-plates or ramps, the over-constrained six-contact base *will* rock and intermittently unload a drive wheel — degrading traction, odometry and stability (worsened by the high CoG). Before field trials this must be resolved.
- **Ranking against other open items.** It sits *below* the two hard geometric conflicts that decide whether the chassis can be built at all — the battery pack aspect ratio vs. the motors, and the caster mount height vs. the 40 mm ground clearance — and below freezing the drivetrain torque (the 33:1 $\rightarrow$ 51:1 ratio question). Resolve those first; schedule the suspension decision immediately after, because choosing a ready C-module *is* the drivetrain decision (it replaces the motor) and therefore should not be left late.

Recommended action: issue an RFQ to ZHLUN and BeUDMKE for their smallest 24 V suspension differential drive wheel (Ø180, ≥ 1.5 m/s, encoder, spring tuned to ~ 65 kg/wheel), and in parallel keep the fabricated rocker (Solution D around the existing Maxon) as the fallback that changes no specification and no dimension.

References

- [1] maxon group. *EC45 flat Ø45 mm brushless, 70 W* — product/datasheet (max. continuous torque 130 mNm, up to 85% efficiency, 24 V winding). <https://www.maxongroup.com/>
- [2] iFixit. *Roomba 650 Wheel Module Disassembly* (identifies the suspension pivot pin). <https://www.ifixit.com/Guide/Roomba+650+wheel+module+Disassembly/97733>

- [3] Sevarg. *Roomba i7 Teardown* (drive module = motor + geartrain + suspension spring, with large travel). <https://www.sevarg.net/2018/12/30/roomba-i7-teardown-with-the-waving-cat/>
- [4] USPTO. *Autonomous mobile robot with a single modular platform*, US 12,128,554 (coil-spring drive units on linear bearings, ≈ 25 mm stroke, parallel suspension).
- [5] USPTO. *Driving module of autonomous mobile robot*, US 12,194,800 (single-pivot rocker drive unit).
- [6] The Planetary Society. *How the Mars rover rocker-bogie suspension works* (“if the left front wheel goes up, the right front wheel is forced down by the same amount”). <https://www.planetary.org/>
- [7] Mobile Industrial Robots. *MiR600 specifications* (ground clearance ≈ 27 mm; traversable floor gap up to ≈ 30 mm). <https://www.mobile-industrial-robots.com/>
- [8] ZHLUN / Guangzhou Wisdom Wheel. *Robotic Drive Units — Electric Differential Wheel* (two-stage spring + PU damping). <https://www.agv-wheel.com/>
- [9] TZBOT. *AGV Drive Wheel — Suspension (differential)*. <https://www.tzbotautomation.com/agv-drive-wheels/differential-drive-wheel/agv-drive-wheel-suspension.html>
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- [12] Bosch Rexroth. *ROKIT Motor — all-in-one wheel drive for AMRs*. <https://www.boschrexroth.com/>
- [13] IEEE/IROS. *Flexible (multilayered) suspension mechanism for a differential-drive mobile robot* (passive spring+damper, self-levelling; for tall, tip-prone robots).